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MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
MID TERM EXAMINATION March 2024

Programme: B. Tech. (Electrical Engineering)

Semester: IV

Course Name: EMEC II

Subject Code: EE-222

Time Duration: 90 Minutes

Maximum Marks (MM): 20

Q.N.	Question		
1.	i.	Why Induction Motor cannot run at synchronous speed?	MM
	ii.	The rotor resistance starting of slip-ring induction motor reduces starting current and increases starting torque. Justify this statement.	CO1
	iii.	What are cogging and crawling in an Induction Motor?	CO2
	iv.	Two 50 Hz induction motors having six and four poles are cumulatively cascaded while the six-pole induction motor is connected to supply mains. Determine the frequencies of the rotor currents and the slips referred to stator field if the set has 'S%' slip.	CO1
2.	a.	What are the fundamental principles of different starters used for starting a squirrel cage induction motor? Give the answer in brief along with the rating range of motors. Which is the most common method and what is its superiority?	CO1
		A 4 pole, 50 Hz, 3 phase Slip ring induction motor, when fully loaded runs with a slip of 'S%'. Determine the value of resistance to be inserted in series per phase in the rotor circuit to reduce the speed by 12% and the new slip. The rotor resistance per phase is 0.2 ohm.	CO2
		Draw the speed torque characteristic of a three-phase induction motor and establish that the maximum torque occurs at a slip $s = R_2/X_2$ and further show that T_{max} is independent of s .	CO3
3.	a.	Determine the ratio of maximum starting torque to full load torque for:	02
	b.	A 3-phase squirrel cage induction motor has a ratio of maximum torque to full load torque as 2.5:1. Determine the ratio of actual starting torque to full load torque for:	03
4.		i. Direct online Starting	CO2
		ii. Star Delta Starting	
		iii. Auto transformer Starting with tapping of 5X%.	CO3
		X is the sum of last two digits of your scholar number, the rotor resistance and standstill reactance per phase are 0.4 ohms and 4 ohms respectively.	
4.		Estimate the stator current, equivalent rotor current, efficiency, output power and its power factor at a slip of 'S%' for a three-phase induction motor having following data per phase: Stator impedance = $(1+j3) \Omega$, Rotor standstill impedance = $(1+j2) \Omega$, No load shunt impedance = $(10+j50) \Omega$, Volts/phase = 250 Volts,	05

Note: In above questions ASSUME the percentage slip (S%) as last digit of your scholar number. For students having last digit 0 in scholar number such as 10, 20, 30..... ASSUME the slip as 10%.

CO: Course Outcome (As per Syllabus of the Course EE 322).